Biology in Human Welfare

Biology is the youngest of the formalised disciplines of natural science, and progress in physics and chemistry proceeded much faster than in biology.

What the last century showed

- applications of physics and chemistry in our daily life also have a higher visibility than those of biology
- however the twentieth century, and certainly the twenty-first century, has demonstrated the utility of biological knowledge in furthering human welfare, be it in the health sector or in agriculture

Where that utility shows

- the discovery of antibiotics, synthetic plant-derived drugs and anaesthetics have changed medical practice on one hand and human health on the other
- life expectancy of human beings has dramatically changed over the years
- agricultural practices, food processing and diagnostics have brought socio-cultural changes in human communities

In this unit

Ch 7 Human Health and Disease

Ch 8 Microbes in Human Welfare

Monkambu Sambasivan Swaminathan, born 1925

- born in August 1925 in Kumbakonam in Tamil Nadu, he did his graduation and post-graduation in Botany from Madras University
- he worked in different capacities in a large number of institutions in India and abroad, and developed his expertise in genetics and plant breeding

The work at IARI

- the School of Cytogenetics and Radiation Research established at the Indian Agricultural Research Institute (IARI) enabled Swaminathan and his team to develop short-duration high-yielding varieties of rice, including scented Basmati
- he is also known for the development of the concept of crop cafeteria, crop scheduling and genetically improving the yield and quality

The Green Revolution, and after

- Swaminathan initiated collaboration with Norman Borlaug, which culminated in the Green Revolution through introduction of Mexican varieties of wheat in India, and this was highly recognised and appreciated
- he is also the initiator of Lab-to-Land, food security and several other environmental programmes
- he has been honoured with the Padma Bhushan and several other prestigious awards, medals and fellowships by institutions of excellence

7.1 From humors to the immune system

The old idea

- health, for a long time, was considered as a state of body and mind where there was a balance of certain humors, which is what early Greeks such as Hippocrates, as well as the Indian Ayurveda system of medicine, asserted
- it was thought that persons with blackbile belonged to a hot personality and would have fevers, an idea arrived at by pure reflective thought
- the discovery of blood circulation by William Harvey using the experimental method, and the demonstration of normal body temperature in persons with blackbile using a thermometer, disproved the good humor hypothesis of health

What biology says instead

- in later years biology stated that the mind influences our immune system through the neural system and the endocrine system, and that our immune system maintains our health
- hence mind and mental state can affect our health

What else affects health

- genetic disorders, that is deficiencies with which a child is born and deficiencies or defects which the child inherits from parents from birth
- infections
- life style, including the food and water we take, the rest and exercise we give to our bodies, and the habits that we have or lack

7.2 Defining health

More than absence of disease

- health does not simply mean absence of disease or physical fitness, and could be defined as a state of complete physical, mental and social well-being
- when people are healthy they are more efficient at work, which increases productivity and brings economic prosperity, while health also increases longevity and reduces infant and maternal mortality
- balanced diet, personal hygiene and regular exercise are very important to maintain good health, and yoga has been practised since time immemorial to achieve physical and mental health
- awareness about diseases and their effect on different bodily functions, vaccination against infectious diseases, proper disposal of wastes, control of vectors and maintenance of hygiene in food and water resources are necessary for achieving good health

Two kinds of disease

- when the functioning of one or more organs or systems of the body is adversely affected, characterised by the appearance of various signs and symptoms, we say that we have a disease
- diseases can be broadly grouped into infectious and non-infectious, those which are easily transmitted from one person to another being called infectious diseases
- infectious diseases are very common and every one of us suffers from these at some time or other, and some of them, such as AIDS, are fatal
- among non-infectious diseases cancer is the major cause of death, and drug and alcohol abuse also affect our health adversely

7.3 Pathogens, and typhoid

What a pathogen is

- a wide range of organisms belonging to bacteria, viruses, fungi, protozoans and helminths could cause diseases in man, and such disease-causing organisms are called pathogens, most parasites therefore being pathogens as they cause harm to the host by living in or on them
- the pathogens can enter our body by various means, multiply and interfere with normal vital activities, resulting in morphological and functional damage, and they have to adapt to life within the environment of the host, so that pathogens entering the gut must know a way of surviving in the stomach at low pH and resisting the various digestive enzymes
- Salmonella typhi is a pathogenic bacterium which causes typhoid fever in human beings, these pathogens generally entering the small intestine through food and water contaminated with them and migrating to other organs through blood
- sustained high fever of 39° to 40°C, weakness, stomach pain, constipation, headache and loss of appetite are some of the common symptoms, and intestinal perforation and death may occur in severe cases

Confirming typhoid, and pneumonia

- typhoid fever could be confirmed by the Widal test, and a classic case in medicine is that of Mary Mallon, nicknamed Typhoid Mary, who was a cook by profession and a typhoid carrier who continued to spread typhoid for several years through the food she prepared
- bacteria such as Streptococcus pneumoniae and Haemophilus influenzae are responsible for pneumonia in humans, which infects the alveoli, the air filled sacs, of the lungs, so that the alveoli get filled with fluid leading to severe problems in respiration
- the symptoms include fever, chills, cough and headache, and in severe cases the lips and finger nails may turn gray to bluish in colour, while a healthy person acquires the infection by inhaling the droplets released by an infected person or even by sharing glasses and utensils with them
- dysentery, plague and diphtheria are some of the other bacterial diseases in man

The common cold

- rhino viruses represent one group of viruses which cause one of the most infectious human ailments, the common cold, infecting the nose and respiratory passage but not the lungs
- the common cold is characterised by nasal congestion and discharge, sore throat, hoarseness, cough, headache and tiredness, which usually last for 3-7 days
- droplets resulting from cough or sneezes of an infected person are either inhaled directly or transmitted through contaminated objects such as pens, books, cups, doorknobs, computer keyboard or mouse, and cause infection in a healthy person

7.4 Malaria

Plasmodium and its two hosts

- Plasmodium, a tiny protozoan, is responsible for malaria, different species, P. vivax, P. malaria and P. falciparum, being responsible for different types, of which malignant malaria caused by Plasmodium falciparum is the most serious and can even be fatal
- Plasmodium enters the human body as sporozoites, the infectious form, through the bite of an infected female Anopheles mosquito, the parasites initially multiplying within the liver cells and then attacking the red blood cells, resulting in their rupture
- the rupture of RBCs is associated with the release of a toxic substance, haemozoin, which is responsible for the chill and high fever recurring every three to four days
- when a female Anopheles mosquito bites an infected person these parasites enter the mosquito's body and undergo further development, multiplying to form sporozoites stored in the salivary glands, so that the malarial parasite requires two hosts, human and mosquito, to complete its life cycle, the female Anopheles being the vector too

Sporozoites entering the human body from the mosquito bite

Multiplication within the liver cells

Attack on the red blood cells, and their rupture

Gametocytes taken up when a mosquito bites again

Development within the mosquito, back to sporozoites

Figure 7.A Stages in the life cycle of Plasmodium

7.5 Amoebiasis

Entamoeba histolytica

- it is a protozoan parasite in the large intestine of human which causes amoebiasis, that is amoebic dysentery
- symptoms of this disease include constipation, abdominal pain and cramps, and stools with excess mucous and blood clots
- houseflies act as mechanical carriers and serve to transmit the parasite from the faeces of an infected person to food and food products, thereby contaminating them
- drinking water and food contaminated by faecal matter are the main source of infection

Chronic inflammation of the lymphatic vessels

Usually in the lower limbs

One limb grossly swollen, hence elephantiasis

Figure 7.B Diagram showing inflammation in one of the lower limbs due to elephantiasis

Dry, scaly lesions on the skin

A ring-shaped affected area

Accompanied by intense itching

Figure 7.C Diagram showing ringworm affected area of the skin

7.6 Helminths and fungi

Ascaris and Wuchereria

- Ascaris, the common round worm, and Wuchereria, the filarial worm, are some of the helminths known to be pathogenic to man
- Ascaris, an intestinal parasite, causes ascariasis, whose symptoms include internal bleeding, muscular pain, fever, anemia and blockage of the intestinal passage, the eggs of the parasite being excreted along with the faeces of infected persons, which contaminate soil, water and plants, so a healthy person acquires the infection through contaminated water, vegetables and fruits
- Wuchereria, that is W. bancrofti and W. malayi, the filarial worms, cause a slowly developing chronic inflammation of the organs in which they live for many years, usually the lymphatic vessels of the lower limbs, and the disease is called elephantiasis or filariasis, the genital organs also often being affected, resulting in gross deformities, and the pathogens are transmitted to a healthy person through the bite by the female mosquito vectors

Ringworm

- many fungi belonging to the genera Microsporum, Trichophyton and Epidermophyton are responsible for ringworms, one of the most common infectious diseases in man
- appearance of dry, scaly lesions on various parts of the body such as skin, nails and scalp are the main symptoms, and these lesions are accompanied by intense itching
- heat and moisture help these fungi to grow, which makes them thrive in skin folds such as those in the groin or between the toes, and ringworms are generally acquired from soil or by using towels, clothes or even the comb of infected individuals

Hygiene

- maintenance of personal and public hygiene is very important for prevention and control of many infectious diseases
- measures for personal hygiene include keeping the body clean and consumption of clean drinking water, food, vegetables and fruits
- public hygiene includes proper disposal of waste and excreta, periodic cleaning and disinfection of water reservoirs, pools, cesspools and tanks, and observing standard practices of hygiene in public catering, measures particularly essential where the infectious agents are transmitted through food and water, such as typhoid, amoebiasis and ascariasis

Air-borne and vector-borne

- in cases of air-borne diseases such as pneumonia and common cold, in addition to the above measures, close contact with infected persons or their belongings should be avoided
- for diseases such as malaria and filariasis that are transmitted through insect vectors, the most important measure is to control or eliminate the vectors and their breeding places
- this can be achieved by avoiding stagnation of water in and around residential areas, regular cleaning of household coolers, use of mosquito nets, introducing fishes such as Gambusia in ponds that feed on mosquito larvae, and spraying of insecticides in ditches, drainage areas and swamps, while doors and windows should be provided with wire mesh to prevent the entry of mosquitoes
- such precautions have become more important especially in the light of recent widespread incidences of the vector-borne Aedes mosquito diseases such as dengue and chikungunya in many parts of India

What biology has armed us with

- the use of vaccines and immunisation programmes have enabled us to completely eradicate a deadly disease such as smallpox
- a large number of other infectious diseases such as polio, diphtheria, pneumonia and tetanus have been controlled to a large extent by the use of vaccines
- biotechnology is at the verge of making available newer and safer vaccines, and the discovery of antibiotics and various other drugs has also enabled us to effectively treat infectious diseases

7.7 Innate immunity

The first two barriers

- every day we are exposed to a large number of infectious agents, however only a few of these exposures result in disease, because the body is able to defend itself from most of these foreign agents, and this overall ability of the host to fight the disease-causing organisms, conferred by the immune system, is called immunity, which is of two types, innate and acquired
- innate immunity is a non-specific type of defence present at the time of birth, accomplished by providing different types of barrier to the entry of foreign agents, and it consists of four types of barrier
- physical barriers: the skin on our body is the main barrier which prevents entry of micro-organisms, and the mucus coating of the epithelium lining the respiratory, gastrointestinal and urogenital tracts also helps in trapping microbes entering our body
- physiological barriers: acid in the stomach, saliva in the mouth and tears from the eyes all prevent microbial growth

Four peptide chains in each antibody molecule

Two small light chains

Two longer heavy chains

Hence an antibody represented as H2L2

Figure 7.D Structure of an antibody molecule

The other two barriers

- cellular barriers: certain types of leukocytes of our body, such as polymorpho-nuclear leukocytes, that is neutrophils, and monocytes, and natural killer cells, a type of lymphocyte, in the blood, as well as macrophages in tissues, can phagocytose and destroy microbes
- cytokine barriers: virus-infected cells secrete proteins called interferons which protect non-infected cells from further viral infection

7.8 Acquired immunity

Memory

- acquired immunity is pathogen specific and is characterised by memory, which means that when our body encounters a pathogen for the first time it produces a response called the primary response, which is of low intensity
- a subsequent encounter with the same pathogen elicits a highly intensified secondary or anamnestic response, ascribed to the fact that our body appears to have memory of the first encounter
- the primary and secondary immune responses are carried out with the help of two special types of lymphocyte present in our blood, the B-lymphocytes and the T-lymphocytes

Antibodies

- the B-lymphocytes produce an army of proteins in response to pathogens into our blood to fight with them, and these proteins are called antibodies, while the T-cells themselves do not secrete antibodies but help B cells to produce them
- each antibody molecule has four peptide chains, two small ones called light chains and two longer ones called heavy chains, hence an antibody is represented as H2L2
- different types of antibody are produced in our body, IgA, IgM, IgE and IgG being some of them

7.9 The two arms of acquired immunity

Humoral and cell-mediated

- because antibodies are found in the blood the response is also called the humoral immune response, one of the two types of our acquired immune response, the antibody mediated one
- the second type is called the cell-mediated immune response or cell-mediated immunity (CMI), and the T-lymphocytes mediate CMI

Why grafts are rejected

- very often, when some human organs such as heart, eye, liver or kidney fail to function satisfactorily, transplantation is the only remedy to enable the patient to live a normal life, and then a search begins to find a suitable donor
- grafts from just any source, an animal, another primate, or any human being, cannot be made, since the grafts would be rejected sooner or later
- tissue matching and blood group matching are essential before undertaking any graft or transplant, and even after this the patient has to take immuno-suppresants all his or her life
- the body is able to differentiate self and nonself, and the cell-mediated immune response is responsible for the graft rejection

7.10 Active and passive immunity

Made in the host, or given ready-made

- when a host is exposed to antigens, which may be in the form of living or dead microbes or other proteins, antibodies are produced in the host body, and this type of immunity is called active immunity, which is slow and takes time to give its full effective response
- injecting the microbes deliberately during immunisation, or infectious organisms gaining access into the body during natural infection, induce active immunity
- when ready-made antibodies are directly given to protect the body against foreign agents it is called passive immunity, and the yellowish fluid colostrum secreted by the mother during the initial days of lactation has abundant antibodies of the IgA type to protect the infant
- the foetus also receives some antibodies from the mother through the placenta during pregnancy, and these are some examples of passive immunity

7.11 Vaccination and immunisation

The principle

- the principle of immunisation or vaccination is based on the property of memory of the immune system, a preparation of antigenic proteins of a pathogen, or an inactivated or weakened pathogen, being introduced into the body as a vaccine
- the antibodies produced in the body against these antigens would neutralise the pathogenic agents during actual infection, and the vaccines also generate memory B and T-cells that recognise the pathogen quickly on subsequent exposure and overwhelm the invaders with a massive production of antibodies
- if a person is infected with some deadly microbe to which a quick immune response is required, as in tetanus, we need to directly inject the preformed antibodies, or antitoxin, and even in cases of snakebite the injection given to the patient contains preformed antibodies against the snake venom, this type of immunisation being called passive immunisation
- recombinant DNA technology has allowed the production of antigenic polypeptides of pathogen in bacteria or yeast, and vaccines produced using this approach allow large scale production and hence greater availability for immunisation, for example the hepatitis B vaccine produced from yeast

7.12 Allergies and auto immunity

Allergy

- the exaggerated response of the immune system to certain antigens present in the environment is called allergy, and the substances to which such an immune response is produced are called allergens, the antibodies produced to these being of the IgE type
- common examples of allergens are mites in dust, pollens and animal dander, and symptoms of allergic reactions include sneezing, watery eyes, running nose and difficulty in breathing, allergy being due to the release of chemicals such as histamine and serotonin from the mast cells
- for determining the cause of allergy the patient is exposed to or injected with very small doses of possible allergens and the reactions studied, while the use of drugs such as anti-histamine, adrenalin and steroids quickly reduce the symptoms
- modern-day life style has somehow resulted in lowering of immunity and more sensitivity to allergens, more and more children in metro cities of India suffering from allergies and asthma, which could be because of the protected environment provided early in life

Auto immunity

- memory-based acquired immunity evolved in higher vertebrates based on the ability to differentiate foreign organisms from self-cells, and while we still do not understand the basis of this, two corollaries have to be understood
- one, higher vertebrates can distinguish foreign molecules as well as foreign organisms, and most of experimental immunology deals with this aspect
- two, sometimes, due to genetic and other unknown reasons, the body attacks self-cells, which results in damage to the body and is called auto-immune disease, rheumatoid arthritis, which affects many people in our society, being one

Bone marrow and thymus, the primary lymphoid organs

Spleen, a large bean-shaped organ

Lymph nodes along the lymphatic system

Tonsils, Peyer's patches and appendix among the secondary organs

Figure 7.E Diagrammatic representation of Lymph nodes

7.13 The immune system in the body

What it consists of

- the human immune system consists of lymphoid organs, tissues, cells and soluble molecules such as antibodies, and is unique in the sense that it recognises foreign antigens, responds to these and remembers them
- the immune system also plays an important role in allergic reactions, auto-immune diseases and organ transplantation
- lymphoid organs are the organs where origin, maturation and proliferation of lymphocytes occur, the primary lymphoid organs being bone marrow and thymus, where immature lymphocytes differentiate into antigen-sensitive lymphocytes
- after maturation the lymphocytes migrate to secondary lymphoid organs such as spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix, which provide the sites for interaction of lymphocytes with the antigen, after which they proliferate to become effector cells

Each organ in turn

- the bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced
- the thymus is a lobed organ located near the heart and beneath the breastbone, quite large at the time of birth but reducing in size with age until by puberty it is very small, and both bone-marrow and thymus provide micro-environments for the development and maturation of T-lymphocytes
- the spleen is a large bean-shaped organ, mainly containing lymphocytes and phagocytes, acting as a filter of the blood by trapping blood-borne micro-organisms, and it also has a large reservoir of erythrocytes
- the lymph nodes are small solid structures located at different points along the lymphatic system, serving to trap the micro-organisms or other antigens which happen to get into the lymph and tissue fluid, and antigens trapped there are responsible for the activation of the lymphocytes present and cause the immune response

An envelope enclosing the RNA genome

Reverse transcriptase making viral DNA from that RNA

Viral DNA incorporated into the host cell DNA

The infected cell directed to produce virus particles

Figure 7.F Replication of retrovirus

7.14 AIDS

What the name means

- there is lymphoid tissue also located within the lining of the major respiratory, digestive and urogenital tracts, called mucosa-associated lymphoid tissue (MALT), which constitutes about 50 per cent of the lymphoid tissue in the human body
- the word AIDS stands for Acquired Immuno Deficiency Syndrome, meaning deficiency of immune system acquired during the lifetime of an individual, indicating that it is not a congenital disease, while syndrome means a group of symptoms
- AIDS was first reported in 1981 and in the last twenty-five years or so has spread all over the world, killing more than 25 million persons

How it is transmitted

- AIDS is caused by the Human Immuno deficiency Virus (HIV), a member of a group of viruses called retrovirus, which have an envelope enclosing the RNA genome
- transmission generally occurs by sexual contact with an infected person, by transfusion of contaminated blood and blood products, by sharing infected needles as in the case of intravenous drug abusers, and from an infected mother to her child through the placenta
- so people at high risk include individuals who have multiple sexual partners, drug addicts who take drugs intravenously, individuals who require repeated blood transfusions, and children born to an HIV infected mother
- HIV and AIDS are not spread by mere touch or physical contact but only through body fluids, so it is imperative for physical and psychological well-being that infected persons are not isolated from family and society, and there is always a time-lag between the infection and appearance of AIDS symptoms, which may vary from a few months to many years, usually 5-10 years

7.15 How the virus acts

Inside the body

- after getting into the body the virus enters macrophages, where the RNA genome of the virus replicates to form viral DNA with the help of the enzyme reverse transcriptase, and this viral DNA gets incorporated into the host cell's DNA and directs the infected cells to produce virus particles, the macrophages continuing to produce virus and so acting like an HIV factory
- simultaneously HIV enters helper T-lymphocytes, replicates and produces progeny viruses, which are released in the blood and attack other helper T-lymphocytes, and this is repeated, leading to a progressive decrease in the number of helper T-lymphocytes in the body
- during this period the person suffers from bouts of fever, diarrhoea and weight loss, and due to the decrease in helper T lymphocytes starts suffering from infections that could have been otherwise overcome, such as those due to bacteria especially Mycobacterium, viruses, fungi and even parasites such as Toxoplasma, so that the patient becomes so immuno-deficient that he or she is unable to protect against these infections
- a widely used diagnostic test for AIDS is enzyme linked immuno-sorbent assay (ELISA), and treatment with anti-retroviral drugs is only partially effective, prolonging the life of the patient but not preventing death, which is inevitable

7.16 Prevention of AIDS

Prevention is the only option

- as AIDS has no cure, prevention is the best option, and HIV infection more often spreads due to conscious behaviour patterns and is not something that happens inadvertently, though infection in blood transfusion patients and new-borns may take place due to poor monitoring, the only excuse being ignorance
- in our country the National AIDS Control Organisation (NACO) and other non-governmental organisations are doing a lot to educate people about AIDS, and WHO has started a number of programmes to prevent the spreading of HIV infection
- making blood from blood banks safe from HIV, ensuring the use of only disposable needles and syringes in public and private hospitals and clinics, free distribution of condoms, controlling drug abuse, advocating safe sex and promoting regular check-ups for HIV in susceptible populations are some such steps taken up
- infection with HIV or having AIDS is something that should not be hidden, since then the infection may spread to many more people, and infected people need help and sympathy instead of being shunned by society, since unless society recognises it as a problem to be dealt with in a collective manner the chances of wider spread increase manifold

7.17 Cancer

Tumors, benign and malignant

- cancer is one of the most dreaded diseases of human beings and a major cause of death all over the globe, more than a million Indians suffering from cancer and a large number of them dying from it annually
- in our body cell growth and differentiation is highly controlled and regulated, whereas in cancer cells there is breakdown of these regulatory mechanisms, and normal cells show a property called contact inhibition by virtue of which contact with other cells inhibits their uncontrolled growth, which cancer cells appear to have lost, so cancerous cells just continue to divide, giving rise to masses of cells called tumors
- benign tumors normally remain confined to their original location and do not spread to other parts of the body, and cause little damage
- the malignant tumors are a mass of proliferating cells called neoplastic or tumor cells, which grow very rapidly, invading and damaging the surrounding normal tissues and starving the normal cells by competing for vital nutrients, while cells sloughed from such tumors reach distant sites through blood and start a new tumor wherever they get lodged, a property called metastasis, the most feared property of malignant tumors

Causes

- transformation of normal cells into cancerous neoplastic cells may be induced by physical, chemical or biological agents, called carcinogens
- ionising radiations such as X-rays and gamma rays, and non-ionizing radiations such as UV, cause DNA damage leading to neoplastic transformation
- the chemical carcinogens present in tobacco smoke have been identified as a major cause of lung cancer
- cancer causing viruses, called oncogenic viruses, have genes called viral oncogenes, and several genes called cellular oncogenes or proto oncogenes have been identified in normal cells which, when activated under certain conditions, could lead to oncogenic transformation

Detection and diagnosis

- early detection of cancers is essential as it allows the disease to be treated successfully in many cases, and detection is based on biopsy and histopathological studies of the tissue, and blood and bone marrow tests for increased cell counts in the case of leukemias
- in biopsy a piece of the suspected tissue, cut into thin sections, is stained and examined under microscope by a pathologist
- techniques such as radiography, that is the use of X-rays, computed tomography (CT) and magnetic resonance imaging (MRI) are very useful to detect cancers of the internal organs, computed tomography using X-rays to generate a three-dimensional image of the internals of an object and MRI using strong magnetic fields and non-ionising radiations to accurately detect pathological and physiological changes in living tissue

Morphine, extracted from the latex of the poppy

Its ring structure drawn out

Acetylation of it giving heroin

Figure 7.G Chemical structure of Morphine

Papaver somniferum, the opium poppy

The latex of the plant, from which morphine is extracted

The source of heroin, commonly called smack

Figure 7.H Opium poppy

Antibodies, genes and treatment

- antibodies against cancer-specific antigens are also used for detection of certain cancers, and techniques of molecular biology can be applied to detect genes in individuals with inherited susceptibility to certain cancers, identification of which may be very helpful in prevention, such individuals being advised to avoid exposure to particular carcinogens
- the common approaches for treatment are surgery, radiation therapy and immunotherapy, and in radiotherapy tumor cells are irradiated lethally, taking proper care of the normal tissues surrounding the tumor mass
- several chemotherapeutic drugs are used to kill cancerous cells, some of them specific for particular tumors, and the majority of drugs have side effects such as hair loss and anemia, while most cancers are treated by a combination of surgery, radiotherapy and chemotherapy
- tumor cells have been shown to avoid detection and destruction by the immune system, so patients are given substances called biological response modifiers such as α-interferon, which activates their immune system and helps in destroying the tumor

7.18 Drugs and alcohol abuse

Opioids

- surveys and statistics show that use of drugs and alcohol has been on the rise especially among the youth, a cause of concern as it could result in many harmful effects, while proper education and guidance would enable youth to safeguard themselves
- the drugs which are commonly abused are opioids, cannabinoids and coca alkaloids, the majority of these being obtained from flowering plants and some from fungi
- opioids are the drugs which bind to specific opioid receptors present in our central nervous system and gastrointestinal tract, and heroin, commonly called smack, is chemically diacetylmorphine, a white, odourless, bitter crystalline compound obtained by acetylation of morphine, which is extracted from the latex of the poppy plant Papaver somniferum
- generally taken by snorting and injection, heroin is a depressant and slows down body functions

A group of chemicals acting on cannabinoid receptors

Those receptors present principally in the brain

The molecule drawn as a skeleton

Figure 7.I Skeletal structure of cannabinoid molecule

Cannabis sativa, the source plant

Flower tops, leaves and resin used in combination

Giving marijuana, hashish, charas and ganja

Figure 7.K Leaves of Cannabis sativa

Datura, with hallucinogenic properties

Atropa belladona, another such plant

A flowering branch of the first

Figure 7.L Flowering branch of Datura

7.19 Cannabinoids and coca alkaloids

What they are and what they do

- cannabinoids are a group of chemicals which interact with cannabinoid receptors present principally in the brain, natural cannabinoids being obtained from the inflorescences of the plant Cannabis sativa, whose flower tops, leaves and resin are used in various combinations to produce marijuana, hashish, charas and ganja, and generally taken by inhalation and oral ingestion, these are known for their effects on the cardiovascular system
- coca alkaloid or cocaine is obtained from the coca plant Erythroxylum coca, native to South America, and interferes with the transport of the neuro-transmitter dopamine, while cocaine, commonly called coke or crack, is usually snorted and has a potent stimulating action on the central nervous system, producing a sense of euphoria and increased energy, excessive dosage causing hallucinations
- other well-known plants with hallucinogenic properties are Atropa belladona and Datura, and these days cannabinoids are also being abused by some sportspersons
- drugs such as barbiturates, amphetamines, benzodiazepines and other similar drugs, normally used as medicines to help patients cope with mental illnesses such as depression and insomnia, are often abused, while morphine is a very effective sedative and painkiller and is very useful in patients who have undergone surgery, and several plants, fruits and seeds having hallucinogenic properties have been used for hundreds of years in folk-medicine, religious ceremonies and rituals all over the globe, so when these are taken for a purpose other than medicinal use, or in amounts or frequency that impairs physical, physiological or psychological functions, it constitutes drug abuse

7.20 Tobacco

What smoking does

- smoking also paves the way to hard drugs, and tobacco has been used by human beings for more than 400 years, being smoked, chewed or used as snuff
- tobacco contains a large number of chemical substances including nicotine, an alkaloid, and nicotine stimulates the adrenal gland to release adrenaline and nor-adrenaline into blood circulation, both of which raise blood pressure and increase heart rate
- smoking is associated with increased incidence of cancers of lung, urinary bladder and throat, and with bronchitis, emphysema, coronary heart disease and gastric ulcer, while tobacco chewing is associated with increased risk of cancer of the oral cavity
- smoking increases carbon monoxide content in blood and reduces the concentration of haembound oxygen, which causes oxygen deficiency in the body

The warning nobody reads

- when one buys packets of cigarettes one cannot miss the statutory warning present on the packing, which warns against smoking and says how it is injurious to health
- yet smoking is very prevalent in society, both among young and old
- knowing the dangers of smoking and chewing tobacco, and its addictive nature, the youth and old need to avoid these habits, and any addict requires counselling and medical help to get rid of the habit

7.21 Adolescence and abuse

A vulnerable phase

- adolescence means both a period and a process during which a child becomes mature in terms of attitudes and beliefs for effective participation in society, the period between 12-18 years of age being thought of as adolescence, a bridge linking childhood and adulthood, accompanied by several biological and behavioural changes, and thus a very vulnerable phase of mental and psychological development
- curiosity, need for adventure and excitement, and experimentation constitute common causes which motivate youngsters towards drug and alcohol use, a child's natural curiosity motivating experiment, complicated further by effects that might be perceived as benefits
- thus the first use may be out of curiosity or experimentation, but later the child starts using these to escape facing problems, and of late stress from pressures to excel in academics or examinations has played a significant role in persuading the youngsters to try alcohol and drugs
- the perception among youth that it is cool or progressive to smoke, use drugs or alcohol is also in a way a major cause, and television, movies, newspapers and internet also help to promote this perception, while other factors associated with abuse among adolescents are unstable or unsupportive family structures and peer pressure

7.22 Addiction and dependence

Addiction

- because of the perceived benefits, drugs are frequently used repeatedly, and the most important thing one fails to realise is the inherent addictive nature of alcohol and drugs
- addiction is a psychological attachment to certain effects, such as euphoria and a temporary feeling of well-being, associated with drugs and alcohol, which drive people to take them even when these are not needed, or even when their use becomes self-destructive
- with repeated use the tolerance level of the receptors present in our body increases, so the receptors respond only to higher doses, leading to greater intake and addiction, though it should be clearly borne in mind that use of these drugs even once can be a fore-runner to addiction
- thus the addictive potential of drugs and alcohol pulls the user into a vicious circle leading to their regular use, from which he or she may not be able to get out, and in the absence of any guidance or counselling the person gets addicted and becomes dependent

Dependence

- dependence is the tendency of the body to manifest a characteristic and unpleasant withdrawal syndrome if a regular dose of drugs or alcohol is abruptly discontinued
- this is characterised by anxiety, shakiness, nausea and sweating, which may be relieved when use is resumed again
- in some cases withdrawal symptoms can be severe and even life threatening and the person may need medical supervision
- dependence leads the patient to ignore all social norms in order to get sufficient funds to satiate his or her needs, which results in many social adjustment problems

7.23 Effects of abuse

The immediate effects

- the immediate adverse effects of drugs and alcohol abuse are manifested in the form of reckless behaviour, vandalism and violence, while excessive doses may lead to coma and death due to respiratory failure, heart failure or cerebral hemorrhage, and a combination of drugs or their intake along with alcohol generally results in overdosing and even deaths
- the most common warning signs among youth include a drop in academic performance, unexplained absence from school or college, lack of interest in personal hygiene, withdrawal, isolation, depression, fatigue, aggressive and rebellious behaviour, deteriorating relationships with family and friends, loss of interest in hobbies, change in sleeping and eating habits, and fluctuations in weight and appetite
- there may even be some far-reaching implications, since if an abuser is unable to get money to buy drugs or alcohol he or she may turn to stealing, and the adverse effects are not restricted to the person using them, an addict at times becoming the cause of mental and financial distress to the entire family and friends

The longer-term harm

- those who take drugs intravenously are much more likely to acquire serious infections such as AIDS and hepatitis B, the viruses responsible being transferred from one person to another by sharing of infected needles and syringes, and both infections are chronic and ultimately fatal and can be transmitted through sexual contact or infected blood
- the use of alcohol during adolescence may also have long-term effects and could lead to heavy drinking in adulthood, while the chronic use of drugs and alcohol damages the nervous system and liver, causing cirrhosis, and use during pregnancy is also known to adversely affect the foetus

Misuse in sport

- another misuse is what certain sportspersons do to enhance their performance, misusing narcotic analgesics, anabolic steroids, diuretics and certain hormones to increase muscle strength and bulk, to promote aggressiveness and so to increase athletic performance
- the side-effects of anabolic steroids in females include masculinisation, increased aggressiveness, mood swings, depression, abnormal menstrual cycles, excessive hair growth on the face and body, enlargement of clitoris and deepening of voice
- in males they include acne, increased aggressiveness, mood swings, depression, reduction of size of the testicles, decreased sperm production, potential for kidney and liver dysfunction, breast enlargement, premature baldness and enlargement of the prostate gland
- these effects may be permanent with prolonged use, and in the adolescent male or female severe facial and body acne, and premature closure of the growth centres of the long bones, may result in stunted growth

7.24 Prevention and control

Where prevention starts

- the age-old adage that prevention is better than cure holds true here also, and it is true that habits such as smoking, taking drug or alcohol are more likely to be taken up at a young age, more during adolescence
- hence it is best to identify the situations that may push an adolescent towards use of drugs or alcohol, and to take remedial measures well in time, in which regard the parents and the teachers have a special responsibility, parenting that combines high levels of nurturance with consistent discipline having been associated with lowered risk of substance abuse
- avoid undue peer pressure: every child has his or her own choice and personality, which should be respected and nurtured, and a child should not be pushed unduly to perform beyond his or her threshold limits, be it studies, sports or other activities

7.25 The remaining measures

Education, help and vigilance

- education and counselling: educating and counselling the child to face problems and stresses, and to accept disappointments and failures as a part of life, while it would also be worthwhile to channelise the child's energy into healthy pursuits such as sports, reading, music, yoga and other extracurricular activities
- seeking help from parents and peers: help from parents and peers should be sought immediately so that they can guide appropriately, and help may even be sought from close and trusted friends, which besides getting proper advice would help the young to vent their feelings of anxiety and guilt
- looking for danger signs: alert parents and teachers need to look for and identify the danger signs, and even friends, if they find someone using drugs or alcohol, should not hesitate to bring this to the notice of parents or teacher in the best interests of the person concerned, so that appropriate measures can diagnose the malady and the underlying causes and initiate proper remedial steps or treatment
- seeking professional and medical help: a lot of help is available in the form of highly qualified psychologists and psychiatrists, and de-addiction and rehabilitation programmes, with which the affected individual, with sufficient effort and will power, can get rid of the problem completely and lead a perfectly normal and healthy life

7.26 Summary

Health and infection

- health is not just the absence of disease, it is a state of complete physical, mental, social and psychological well-being
- diseases such as typhoid, cholera, pneumonia, fungal infections of skin, malaria and many others are a major cause of distress to human beings, and vector-borne diseases such as malaria, especially one caused by Plasmodium falciparum, if not treated may prove fatal
- besides personal cleanliness and hygiene, public health measures such as proper disposal of waste, decontamination of drinking water, control of vectors such as mosquitoes and immunisation are very helpful in preventing these diseases

7.27 Summary, continued

Immunity, AIDS, cancer and abuse

- our immune system plays the major role in preventing these diseases when we are exposed to disease-causing agents, the innate defences such as skin, mucous membranes, antimicrobial substances present in our tears and saliva, and the phagocytic cells helping to block the entry of pathogens into our body, while if the pathogens succeed in gaining entry, specific antibodies and cells serve to kill them
- the immune system has memory, so on subsequent exposure to the same pathogen the immune response is rapid and more intense, which forms the basis of protection afforded by vaccination and immunisation
- among other diseases, AIDS and cancer kill a large number of individuals worldwide, AIDS caused by the human immuno-deficiency virus being fatal but preventable if certain precautions are taken, while many cancers are curable if detected early and appropriate therapeutic measures are taken
- of late, drug and alcohol abuse among youth and adolescents is becoming another cause of concern, and because of the addictive nature of alcohol and drugs, and their perceived benefits such as relief from stress, a person may try taking these in the face of peer pressure and examination-related and competition-related stresses, and in doing so may get addicted, while education about their harmful effects, counselling and seeking immediate professional and medical help would totally relieve the individual from these evils

7.28 Exercises

From this chapter

1. What are the various public health measures, which you would suggest as safeguard against infectious diseases?
2. In which way has the study of biology helped us to control infectious diseases?
3. How does the transmission of each of the following diseases take place? (a) Amoebiasis (b) Malaria (c) Ascariasis (d) Pneumonia
4. What measure would you take to prevent water-borne diseases?
5. Discuss with your teacher what does 'a suitable gene' means, in the context of DNA vaccines.
6. Name the primary and secondary lymphoid organs.
7. The following are some well-known abbreviations, which have been used in this chapter. Expand each one to its full form: (a) MALT (b) CMI (c) AIDS (d) NACO (e) HIV
8. Differentiate the following and give examples of each: (a) Innate and acquired immunity (b) Active and passive immunity
9. Draw a well-labelled diagram of an antibody molecule.
10. What are the various routes by which transmission of human immuno-deficiency virus takes place?
11. What is the mechanism by which the AIDS virus causes deficiency of immune system of the infected person?
12. How is a cancerous cell different from a normal cell?
13. Explain what is meant by metastasis.
14. List the harmful effects caused by alcohol/drug abuse.
15. Do you think that friends can influence one to take alcohol/drugs? If yes, how may one protect himself/herself from such an influence?
16. Why is that once a person starts taking alcohol or drugs, it is difficult to get rid of this habit? Discuss it with your teacher.
17. In your view what motivates youngsters to take to alcohol or drugs and how can this be avoided?